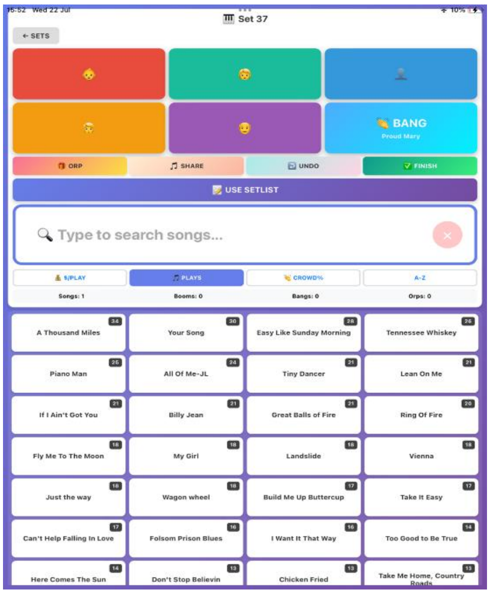
The Stowaway Piano Player Data Project

I've been a piano entertainer for most of my working life. The last 8 years have been mainly dedicated to playing piano and singing in piano bars on Royal Caribbean ships. The contract I'm on now is a different role, travelling piano man or 'stowaway'. The job involves playing piano all over the ship, doing pop-up performances and playing to whoever is walking by, essentially busking on a cruise ship. I'll be on the Utopia of the Seas for 13 weeks. The most profitable sets I do in terms of tips are in the elevator, yes that's right, I play sets in the elevator! People love these performances; they get very excited. These sets provide a unique opportunity for data collection and analyses. People are getting on the elevator, staying for a few minutes (sometimes more), and then they get off. I have about thirty seconds to play something that gets their attention, to get them singing, laughing or just enjoying the ride! Sometimes playing a quick chorus of something I suspect they know and like works, sometimes just playing a showpiece on the piano or improvising some jazz works, sometimes saying something silly, sometimes playing something the kids will like works, sometimes just playing whatever I feel like. Another great thing about the elevator sets is that people tip, this ship is out of Florida where I know from experience people are generous. As I'm interested in data analytics it gave me the idea to turn this job into a project.

The questions I'll be asking, does song choice matter, does crowd reaction correlate to tips earned. I'll also be using any other insights to try to increase the crowd reaction I get in the elevator and increase the amount of tips I get.

The first problem to solve was the method in which I would collect data. For good quality data I need to track when a song gets a tip and when a song gets a crowd response. This was the biggest challenge. My first attempts failed. I tried note taking, I tried taking an audio recording of the script and extracting songs from that with a code word I would say out loud that meant a tip was received. Those methods proved unworkable. I ended up designing a simple app which I enlisted Claude to build. In the app I start a set, this records the start time, then a song is selected from a list and while the song is 'active' one can press a button upon receiving a tip and the tip is recorded and attached to the song. the same for crowd reaction. It's a simple system that requires me to select a song before I play it and then simply attach a crowd reaction or a tip to that song. Later iterations of the app included the approximate age of the guest tipping. I was interested to do a generational analysis – what songs do age brackets tip'. At the end of the set the set is closed and the time is recorded. Then there are some other values entered, total earned in tips during the set is recorded after counting up, the location of where the set is and what port we were in that day, the subjective energy level is also tracked along with footfall level. All of these were designed to gain any insights at the end of the experiment.



This is a picture of the app to the left. Songs are selected from the list and the the buttons at the top record tips or crowd reaction events attached to that song and are exported in the data.

I then ran the app for every set I played for three weeks. I'd set up in the elevator and start a set, track every song I played. How many tips it earned, whether or not I got people singing along- I landed on the rule that if more than one person sings along, the song event earns a crowd reaction in the data.

The data from the experiment is presented below. I ran all these charts initially in SQL as script which I have published on my GitHub. Given the findings and insights were fairly basic I decided against using Power BI and used the results of my SQL scripts to generate some simple charts within Claude.

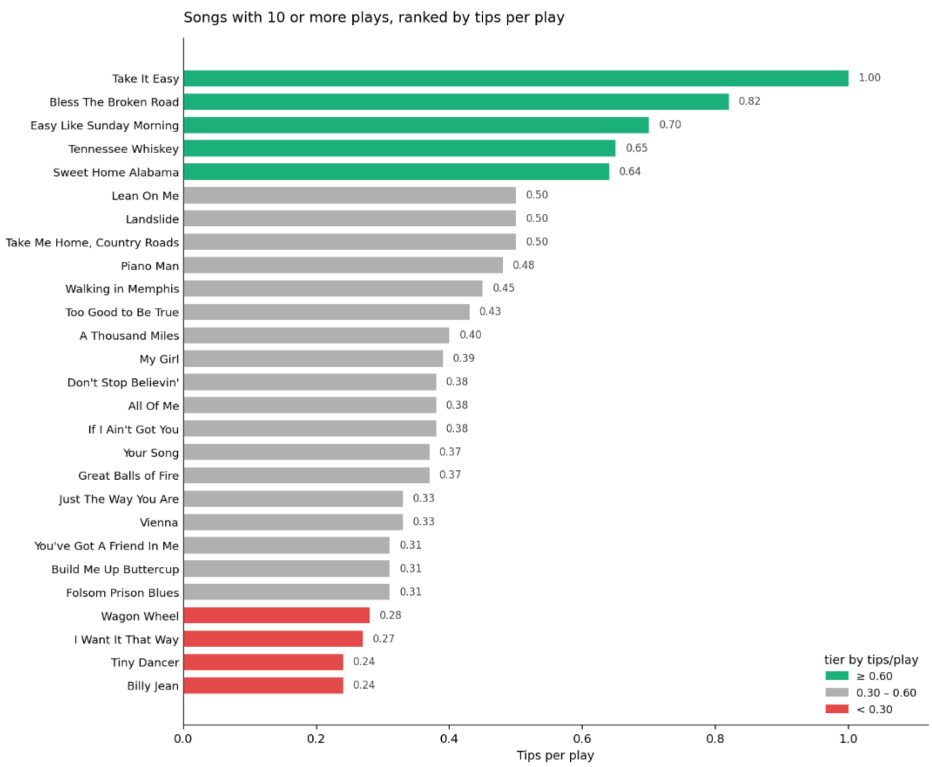
THE DATASET

22 days · 33 sets · 42.4 hours of playing.

THE STUDY	
Sets played	33
Days played	18
Experiment length	22 days
Total hours performed	42.4
Songs played	1,047
Unique songs drawn on	202
EARNINGS	
Total cash	$3,138
Total tips	507
– Tagged to a song	375
– Orphan (song-independent)	132 (26%)
AVERAGES	
Cash per hour played	$74
Cash per set	$95
Cash per song	$3.00
Tips per set	15.4
Songs per set	31.7
Songs per hour	24.7
BASELINES	
Tips per song (all tips)	0.48
Tips per song (tagged only)	0.36

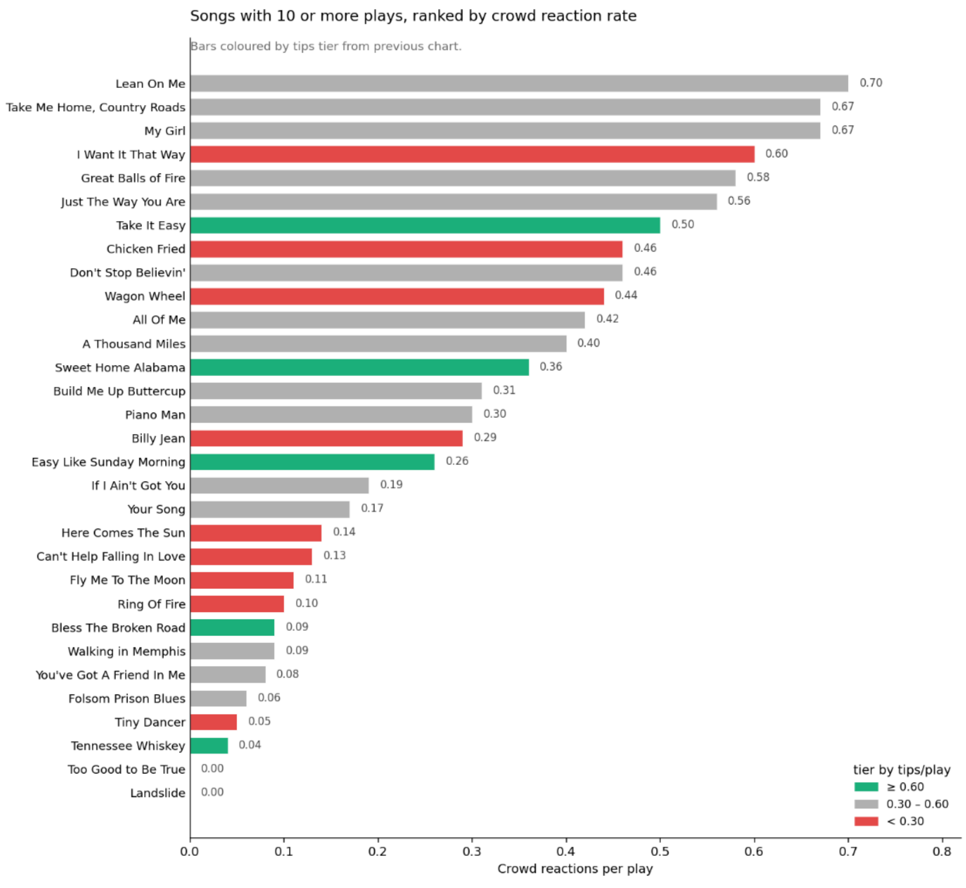
Songs ranked by tips per play

27 songs with 10 or more plays. Green tier earns 0.60+ tips per play, red under 0.30. Baseline for a random play was 0.36. Take It Easy sits at the top with 14 tips in 14 plays; against baseline, that's very unlikely to be chance (p=0.0007). The tier the song sits in is more reliable than its exact rate.



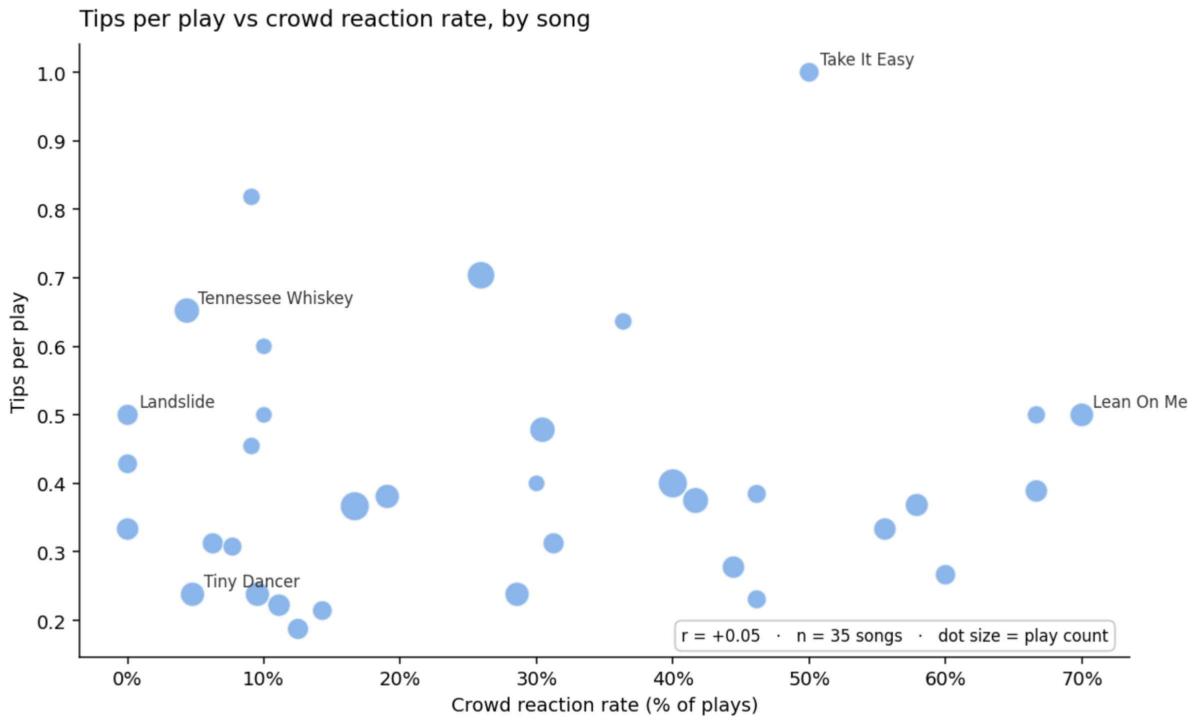
Songs ranked by crowd reaction rate

The same songs re-ranked by how often two or more people sang along. Bars are still coloured by their tips tier — so green scattered through the middle and red near the top means the two rankings don't agree. Lean On Me draws the biggest response but earns mid-tier tips. Tennessee Whiskey and Bless The Broken Road earn top-tier tips almost silently.



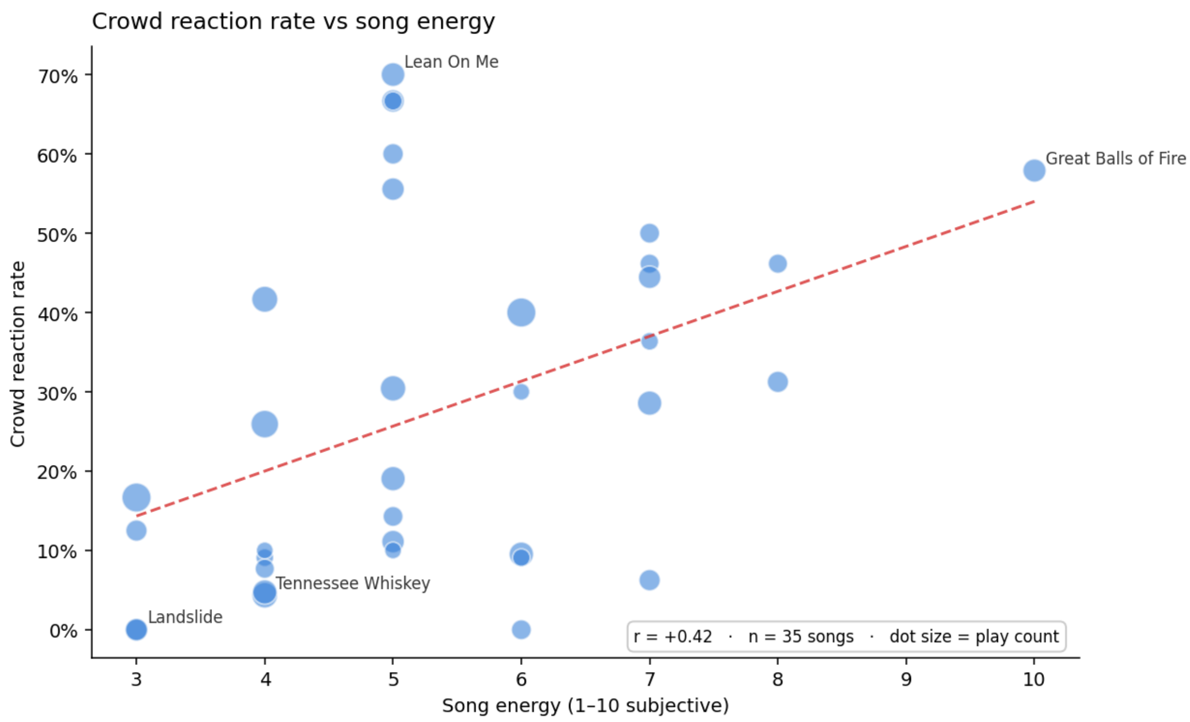
Tips per play vs crowd reaction rate, by song

Each dot is one song, sized by play count. If crowd reaction predicted earnings the cloud would slope. It doesn't (r=0.05 across 35 songs). A play that gets a reaction does tend to earn more in the moment (0.45 vs 0.33 average tips) — but a song's tendency to *produce* reactions doesn't make it a top earner.



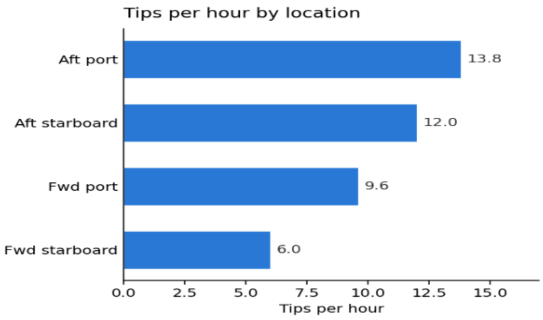
Crowd reaction rate vs song energy

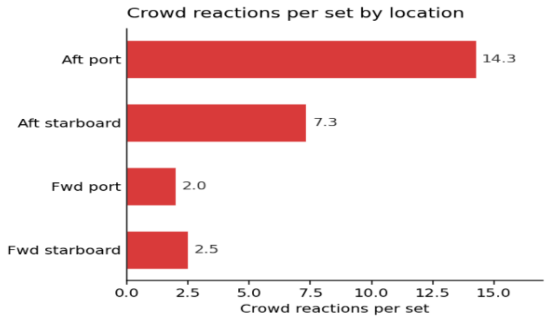
Higher-energy songs draw more crowd reactions. This trend does not carry over to tips — energy predicts singing, not tipping. I scored energy myself, so this is exploratory.



Location

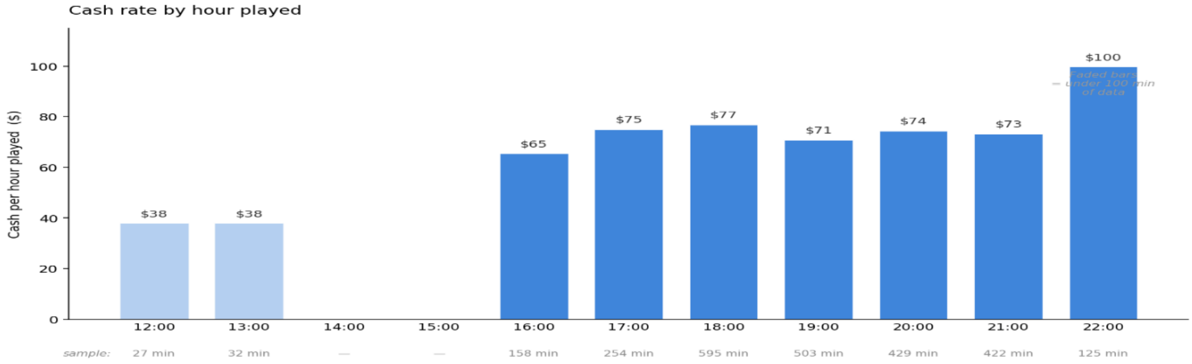
Left: tips per hour by location. Right: crowd reactions per set. Same ranking on both. Aft locations clearly outperform forward ones (12.6 vs 8.4 tips/hour, $p=0.008$). Port trends above starboard on both sides but the sample is too small to be sure. Fwd starboard has just 2 sets — treat as directional. That said, I knew from experience that fwd starboard performed badly which is why it was only used in the experiment a few times.





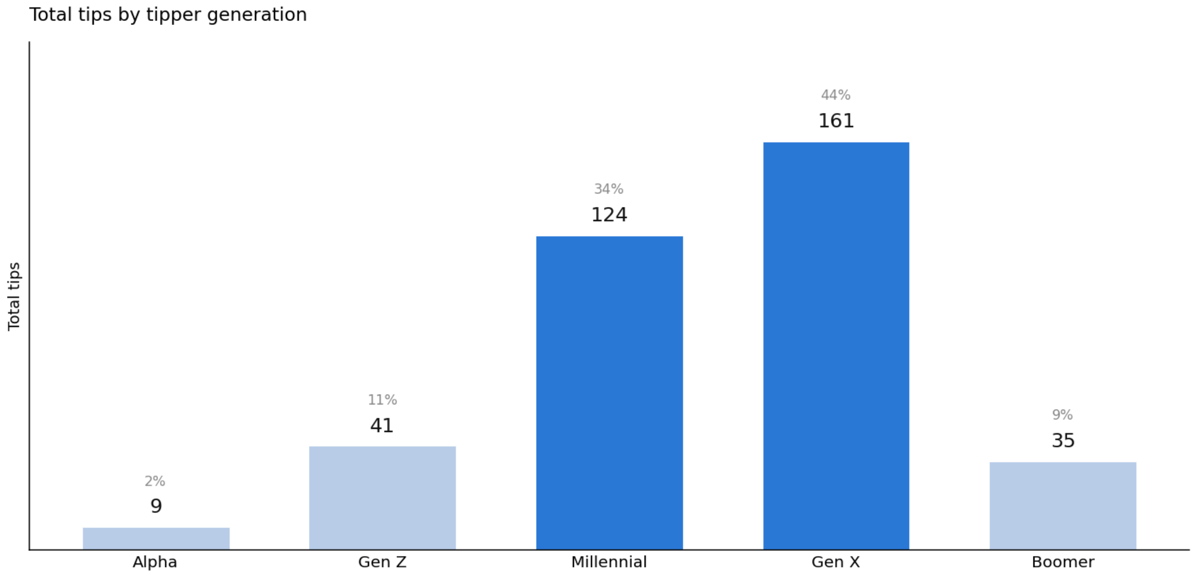
Cash rate by hour played

Midday rates are visibly lower but come from very small samples. From 16:00 onwards the hourly rate sits between \$65–\$77 per hour without a clear trend across the evening. The 22:00 bar looks like a peak but is 125 minutes of data — not enough to say it's a genuine late-evening ramp. My instinct from the months before this experiment was that later sets earn more. This sample is consistent with that direction, though not strong enough on its own to call it proven — a longer collection window with more late-evening sets would settle it.



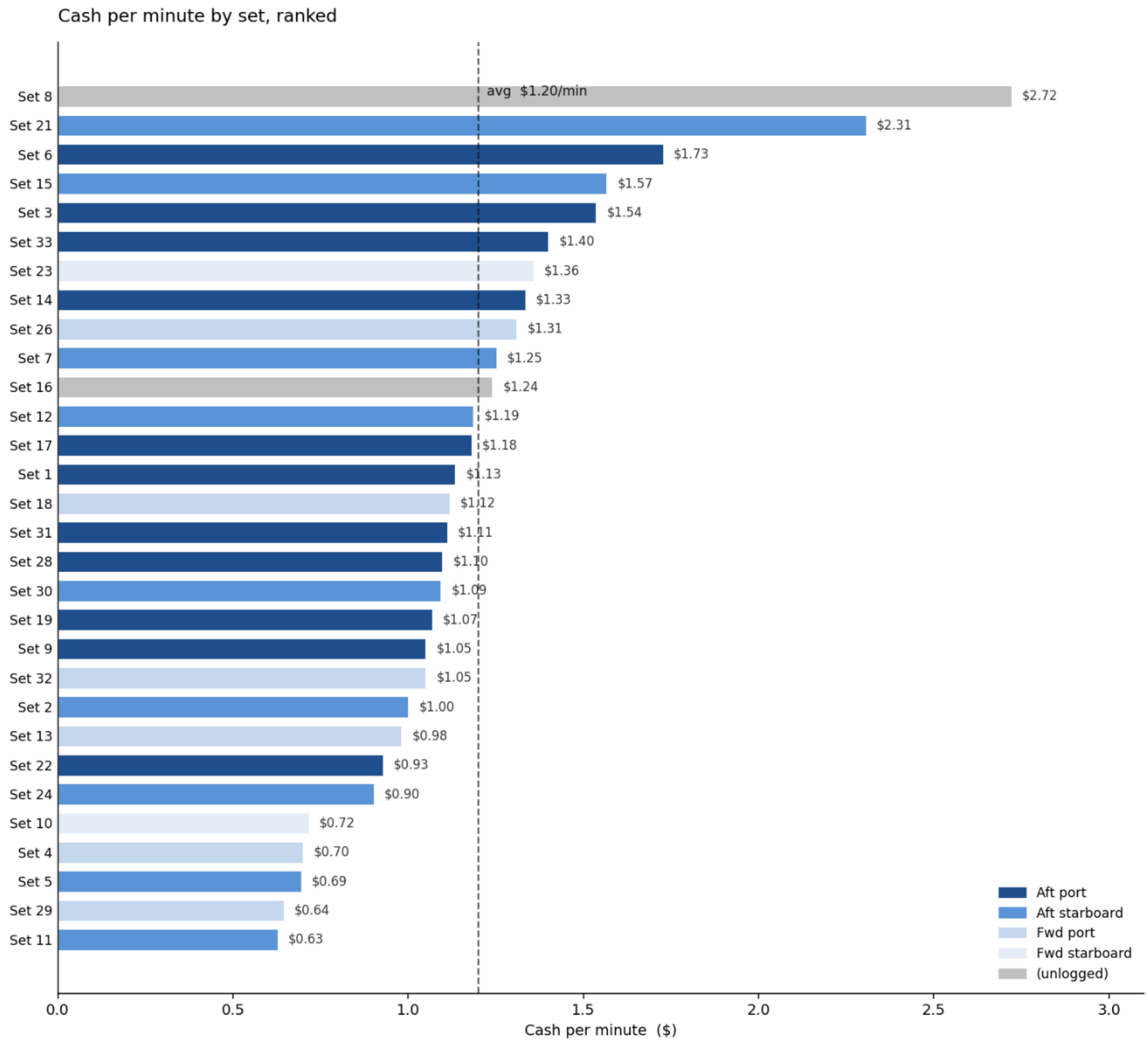
Total tips by tipper generation

Generation of the tipper, I make a guess as to what age bracket they fall in. Gen X and Millennials together account for 77% of tips. I added another dimension to the experiment – does song affect tipping rates for each generation. The answer was no, I didn't go ahead with the full breakdown of this in this presentation but below are the results by generation. Looking back, crowd reaction would have been better to track by generation – clearly relevant and of interest to performers such as myself.



Cash per minute by set, ranked

All 30 completed sets, sorted by cash rate, coloured by location. Dashed line is the average across sets ($1.20/min). Note the range — $0.63 to $2.72 per minute. This scale of set-to-set variance is why smaller sub-effects (side, hour) are hard to prove on 30 sets: a single set can deviate this much on crowd luck alone.



Strategic recommendations.

I've been able to show that some songs are higher earners than others. It also shows that in general, songs that I know to be crowd pleasers are. I can confidently play the high earners more and drop those that have performed badly.

I also checked whether the ship's day port (Nassau, Cococay, Port Canaveral, or a sea day) predicted earnings. It didn't — rates were flat across all four, well within the set-to-set noise.

After years of doing this, nothing is really new to me here. There are a few songs that have performed far better than what I expected and some that I can confidently remove from the set. Songs that work well in a piano bar setting don't translate to top ranked songs in the elevator. Tiny Dancer for example, has proven to be unsingalongable to the cruise ship masses even though I know it to be a classic that piano bar enthusiasts know and love. It's this kind of thing that make this job challenging. Having such a varied range of demographics make it very hard to know what to play. I believe that as the culture of music grows it is increasingly hard to appeal to the masses with classic songs. Young people don't necessarily know songs new or old, old people definitely know old songs but don't always know massive hits that we assume everyone knows. Country folk from the south tend to not like R n B, but sometimes they love it! I could go on….

Knowing this, sticking to the piano bar seems the comfortable option although playing to a sample more representative of the population has been fun.

I would have liked to continue with this experiment and show the recommendations have worked in my favour. Unfortunately I have run out of time and I think the song level data shows the limitations of song choice as a predictor of tip earnings. The problem here is that my intuition in choosing a song for a specific group that I come across may be the very reason the top songs have performed so well. This is a finding in itself – in many cases I should keep doing what I'm doing. If I wanted to test these songs further I could play them more often, to groups of every sort. This would be counter productive though, there is no point playing 'Take It Easy' to a group of 13 year olds. They won't be entertained by it, which is my job, and it won't change the fact that they have no money in their pocket.

If I was to run the experiment again I'd drop song choice altogether. I'd focus on age, time, date and itinerary. I'd track exact times of each tip and count up more regularly so that I could run tighter analysis on when to play, to whom I should focus on and when to avoid. That said, I probably won't be running this experiment again. It's not easy tracking every tip and crowd reaction while playing despite the sleek design of the app. Also, I am moving away from music into a career in data analytics.

The fact that crowd reaction and tips earned don't correlate is perhaps the most interesting finding and goes against what many other piano entertainers that do this job have told me. Playing songs that get the crowd going is exhausting, sometimes fun and completely natural, other times it feels forced. These findings show that I shouldn't worry about that, I have always maintained that as entertainers we should keep expectations low and take each set as it comes. There are just too many variables out of our control. We should play whatever we feel at the time, play the moment, not yesterday's or the best night from last year. These findings show that there are some undoubted hero songs within the piano bar repertoire but that response and tips will come as a result of us doing our best at the time, not following a set of data driven rules. If there are some rules, I'd love to know them and be able to show them but perhaps on this occasion, my instinct is good enough.

This project has been a fun way to join my music career with my ongoing data analyst development.